Connectivity Community of Practice for Applied Internet Measurement

Advancing School Connectivity Measurement

Measurement Lab (M-Lab) × Giga (UNICEF&ITU)

March 11, 2026

Welcome & Agenda!

- 📢 Roll call [5 mins]
- 🎪 Welcome [5 mins]
- 💭 Introducing the Community of Practice (20 mins)
- 💭 Q&A Discussion (20 mins)
- 👪 Close out [5 mins]

Please note: We will record this session and have a transcript!

Welcome & Agenda!

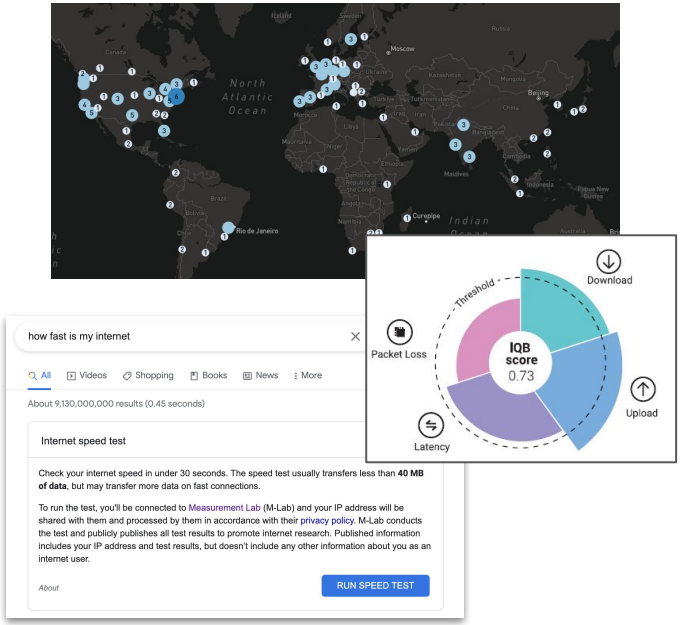
Speakers

- Maria Antonia Bravo – Giga, UNICEF (Digital Inclusion)
- Melissa Newcomb – Measurement Lab
- Pavlos Sermpezis – Measurement Lab

Convening Institutions

About Measurement Lab

- Who we are?
  - M-Lab is an **open** source **non-commercial** project; founded in 2009
  - We maintain a **worldwide server platform**
  - **Researchers** can deploy **open-source experiments**
- What we do?
  - **Measure the Internet** with open-source experiments (e.g., NDT / speed test)
  - Save the data **as open data**
  - Make it universally accessible and useful
  - Research
- Key facts
  - **Platform**: 400+ servers in 40+ countries / 100+ metros
  - **Data**: 4+ million tests per day, total 6+ billion test results



About Giga

Giga: Connecting Every School to the Internet

Giga is a joint initiative of **UNICEF and the ITU** working with governments to connect schools globally.

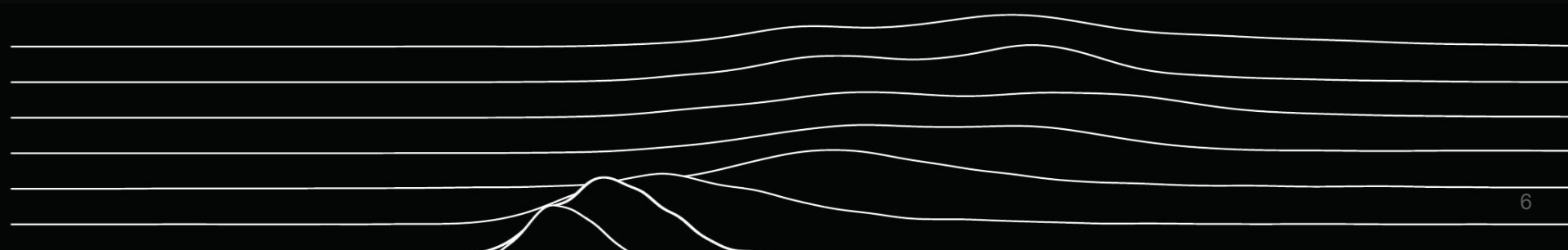
Key focus areas:

- Mapping school connectivity worldwide
- Supporting infrastructure planning
- Financing connectivity deployment
- Providing open data tools for decision-making

Giga works with governments and partners to ensure connectivity leads to:

- Access to learning resources
- Digital inclusion
- Opportunity for young people

Why This Matters



Why This Matters

Connectivity has become foundational infrastructure for education. But globally we still lack consistent ways to answer:

- Are schools actually connected?
- Is the connectivity sufficient for teaching learning? Is it reliable?
- Does the measured performance reflect real student experience? Does it reflect facility-level conditions?

Why this matters

West African Country

1 GB month cap

Exclusive connection to Ministry of Education server for information management system purposes

Analog EMIS backup

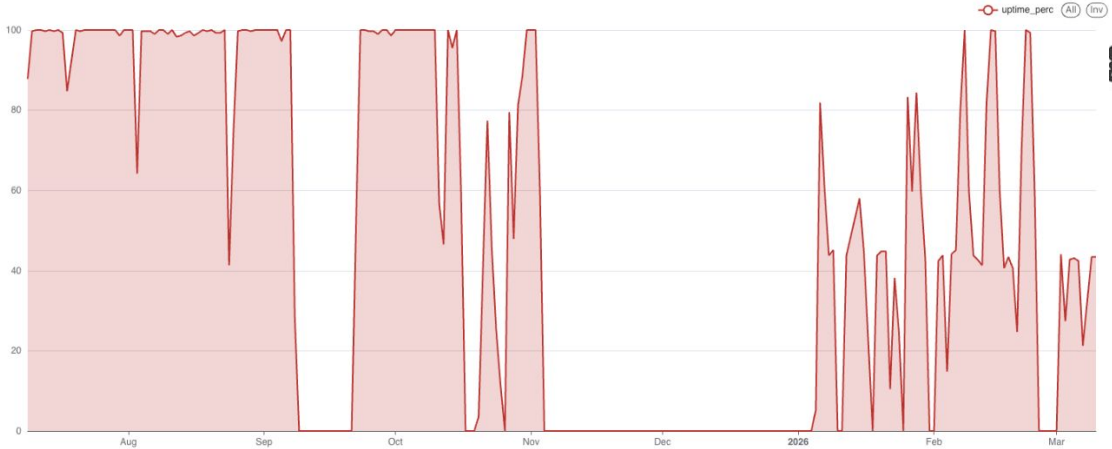
Why this matters

Central Asian School

Well provisioned network + end-user devices

310 pupils

Requirement for weekly IT lessons



Why this matters

Southern African school

VSAT connection

Active end-user measurement

Server location 1,200 km away, crossing two countries

Why this matters

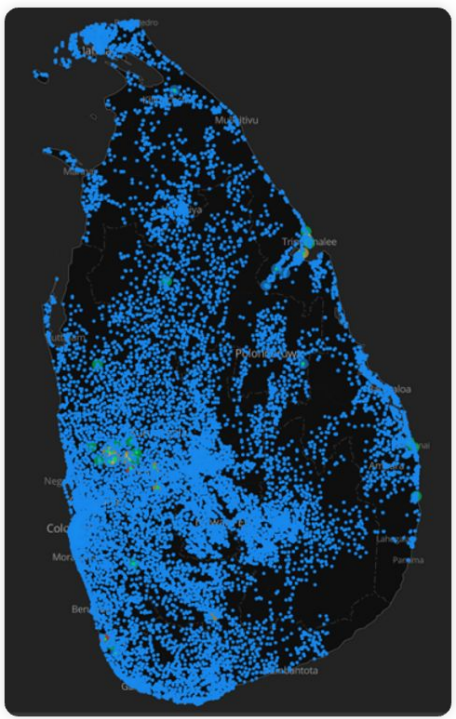
Sri Lanka

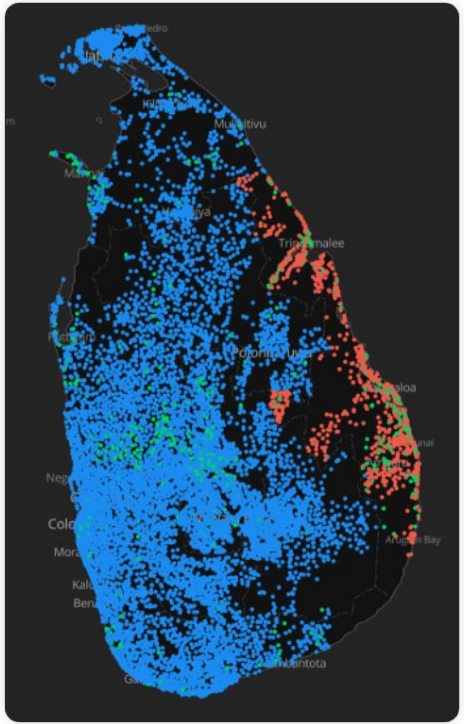
Little baseline data, no connectivity data on school census

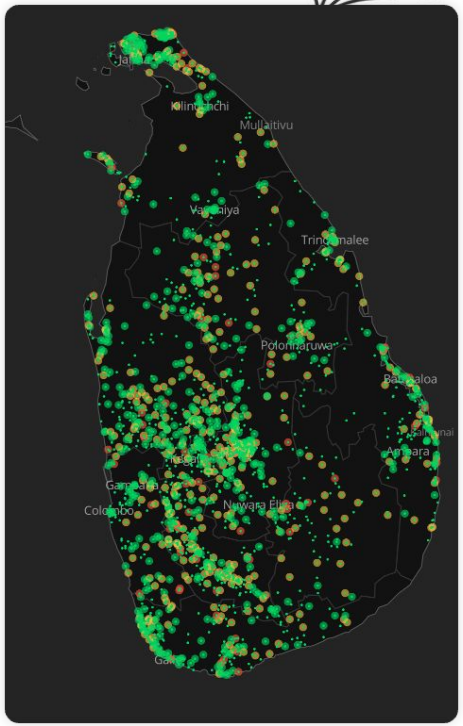
End-user measurements

2024 (left)

2026 (right)







Today:

- Measurement approaches vary by protocol, tooling, vantage point, frequency - results aren't comparable across countries or set-ups
- Governments (and other decision makers) still rely on static, self-reported data (PDFs of NOC dashboards, 'yes-no' EMIS) to make decisions

→ Broad structural gap between what is measurable and what decision makers can use to make decisions

Interpretable

Reproducible

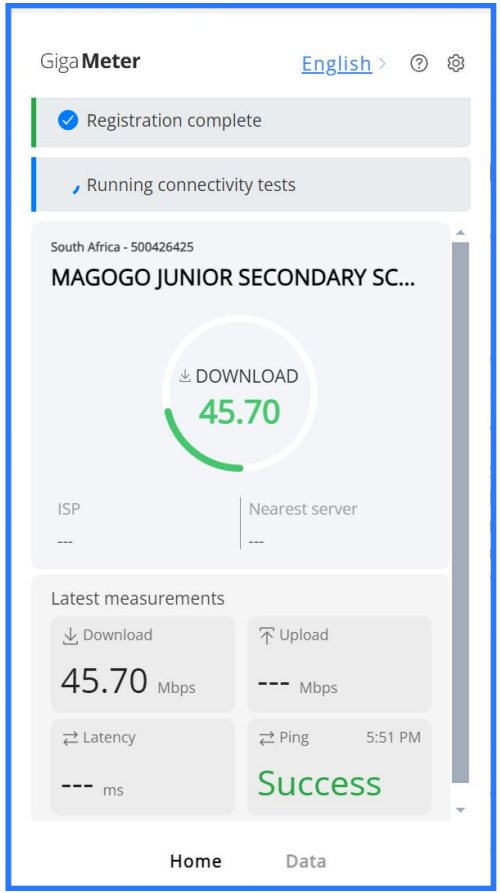
Policy-legible

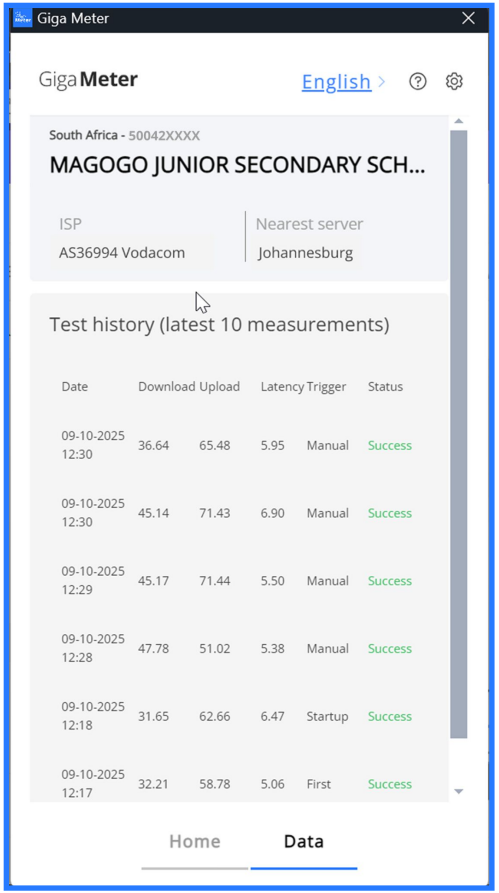
Our work so far:

Roll out of **Giga Meter**, open source measurement tool installed in end-user devices in schools

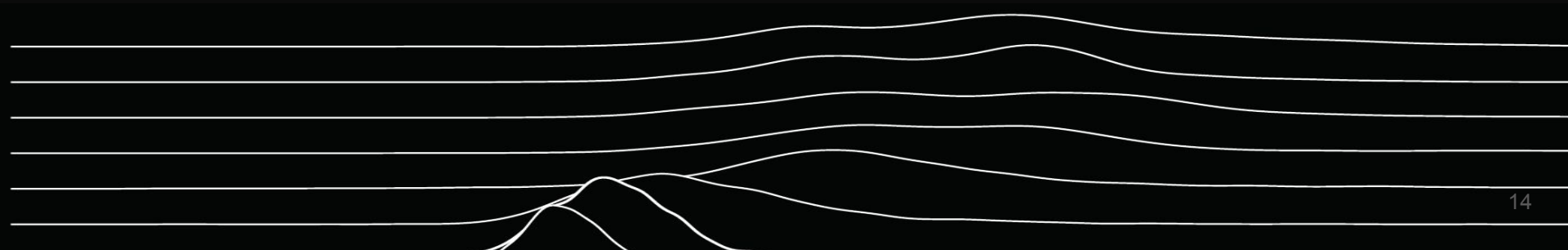
- Scheduler - 4x per day for speed tests (NDT7), every 15 minutes Ping
- M-Lab server fleet (+ soon Cloudflare)
- 6M measurements
- 31 countries
- 14,992 schools

Data available via https://maps.giga.global/docs/explore-api





Introducing the Community of Practice



Introducing the Community of Practice

Connectivity Community of Practice for Applied Internet Measurement

A collaborative forum bringing together:

- Internet measurement researchers
- Network engineers
- Education system practitioners
- Policymakers
- Infrastructure providers
- Digital equity advocates

Together we aim to:

- Strengthen how connectivity is measured in schools
- Translate measurement into usable insights
- Align on practical indicators for meaningful connectivity

Guiding Principles

The CoP is grounded in three principles:

1. Technically Rigorous – Use robust Internet measurement methods and open telemetry.
2. Globally Comparable – Enable cross-country benchmarking and shared indicators.
3. Practical for Real-World Contexts – Measurement approaches must work in resource-constrained environments and underserved regions.

How the Community Will Work

Initial prototype phase (Year 1)

The CoP will:

- Convene regular community sessions
- Form thematic working groups
- Develop shared measurement approaches
- Document research and technical outputs

Governance is initially stewarded by:

- Measurement Lab
- Giga
- Superbloom Design (facilitation)

Introducing the Initial Research Agenda

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Workstreams / Initial areas of exploration:

1. Improving measurement tools
2. Characterization of facility-based connectivity
3. Internet Quality Barometer for Education (“IQB - Edu”)
4. New insights from existing measurement data
5. Open topic - BYOI (Bring Your Own Ideas)!

Research Agenda: Workstream 1

Workstream 1: Improving measurement tools

- Context : Connectivity measurements typically involve speed tests (e.g., NDT7) and ping tests.
- Research questions: *What* other measurements we could consider and *how*?
  - Application layer measurements - e.g., set of domains with education-related content
  - Video streaming - e.g., neubot/dash tests (M-Lab platform)
  - DNS
  - Other?

Research Agenda: Workstream 2

Workstream 2: Characterization of facility-based connectivity

- Context : For each speed test (NDT7), M-Lab also does traceroutes, however, this information has not been leveraged so far for school connectivity.
- Research questions: Traceroutes - what potential do they offer?
  - Analyze traceroutes information to filter data in statistics calculations (e.g., avoid routes through bottlenecks)
  - Use traceroutes information to map school connectivity “ecosystem”: How each school is connected to the outside world? E.g., distance to main ISPs or CDNs, number of (1-hop, 2-hop, N-hop) upstream providers, connectivity to IXPs, reliability/diversity of paths, etc.
  - Other approaches / explorations?

Research Agenda: Workstream 3

Workstream 3: IQB - Edu

- Context : IQB a framework to assess Internet Quality from speed test data. Current design focused on individual Internet users, not facilities.
- Research questions:
  - Design the IQB-Edu framework design (levels: user + facility)?
    Starting with the use case of Moldova
  - Apply the IQB-Edu methodological design (network parameters, weights, etc)
  - What other measurements and/or information would be needed to be included in the IQB-Edu framework (e.g., other type of measurements, measurements to services relevant to school activities)

Research Agenda: Workstream 4

Workstream 4: New insights from existing measurement data

- Context : GigaMeter collects & visualizes aggregated data about school connectivity (e.g., mean throughput or latency)
- Research questions:
  - What other (already collected) metrics are interesting for school connectivity?
  - How to develop diagnostics based on NDT7 data (e.g., timeseries, anomaly detection)?
  - What else do measurements can tell us about school connection quality and performance?

Research Agenda: Workstream 5

Workstream 5: Open topic

- BYOI (Bring Your Own Ideas)! …

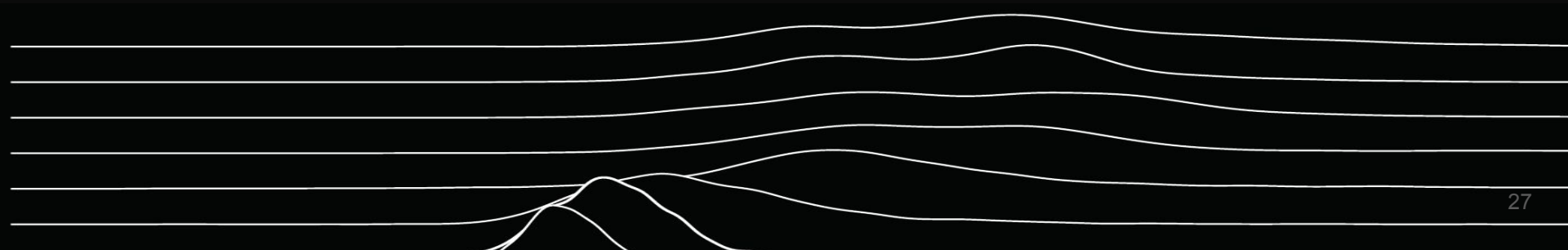
How You Can Participate

How You Can Participate

Ways to engage in the CoP:

- Join community sessions
- Contribute to working groups
- Share measurement approaches or case studies
- Provide feedback on research outputs
- Participate in pilots or experiments
- Participation is open and flexible.
- See more information on governance

What We Hope For From This Community



What We Hope For From This Community

Today's launch is the beginning. We want to learn from you:

- What measurement challenges are you facing?
  - e.g., lack of data / deployments / tools, data quality or reliability
- What indicators matter most for schools?
  - e.g., network measurements vs use cases
- Where do current tools fall short?
- What would you like to see as outputs?
  - e.g., reports, visualizations, dashboards, code/software
- What should the first working groups focus on?
- How to join forces with other related efforts?

Your input will shape the CoP's first-year agenda.

Thank you!